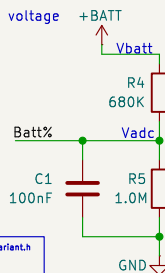


Battery voltage divider for measuring voltage

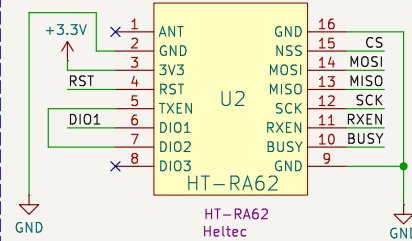
$R4 = 470$
 $R5 = 680$
 # Voltage Division Equation
 $V_{adc} = V_{batt} * (R5 / (R4 + R5))$
 # Voltage Vbatt --> Vadc ratio
 $ADC_Division = V_{adc} / V_{batt} = (R5 / (R4 + R5))$
 $R5 / (R4 + R5) = 0.591304347826087$
 # Voltage Vadc --> Vbatt ratio
 $ADC_Multiplier = V_{adc} / V_{batt} = (R4 + R5) / R5$
 $(R4 + R5) / R5 = 1.6913764705882353$

C1 used for getting more stable and accurate readings

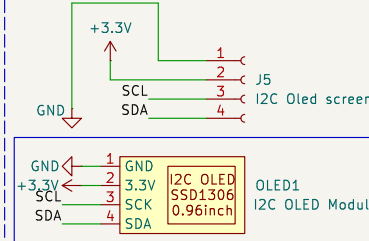
firmware/variants/nrf52840/dy/nrf52_promicro_diy_txco/variant.h
 #define VBAT_DIVIDER_COMP (1.73)
 #define ADC_MULTIPLIER VBAT_DIVIDER_COMP



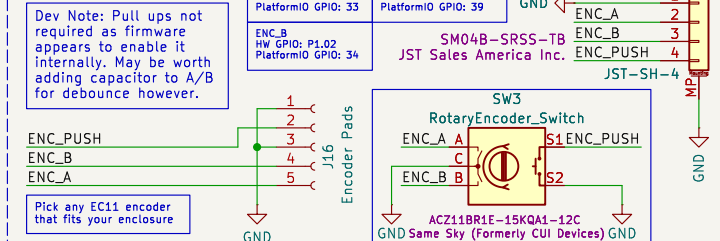
HT-RA62 / RA-01SH



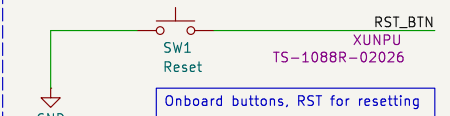
Pins for connecting an I2C OLED screen



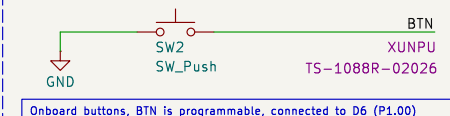
Pads for connecting encoder



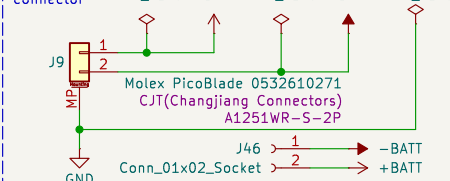
Onboard buttons RST



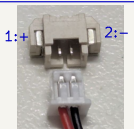
Onboard buttons BTN



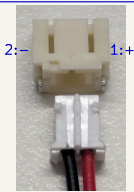
Battery connector



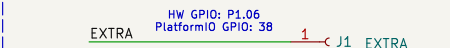
Molex PicoBlade 2-Pin
 - MPC: Molex, MPN: 0532610271
 - MPC: CJT (Changjiang Connectors), MPN: A1251WR-S-2P
 - Pitch: 1.25mm
 - Used By: HelTec LoRa 32
 - This polarity is also common for lipos in Aliexpress



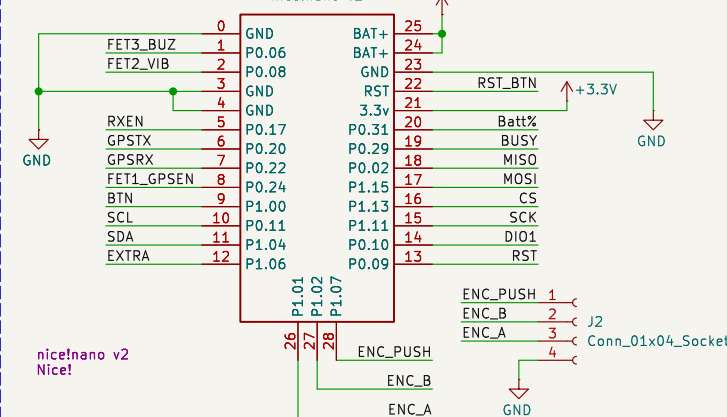
JST PH 2-Pin
 - MPN: S2B-PH-SM4-TB
 - Pitch: 2.00mm
 - Used By: RAK19007 WisBlock
 - This polarity is also common for lipos in Aliexpress



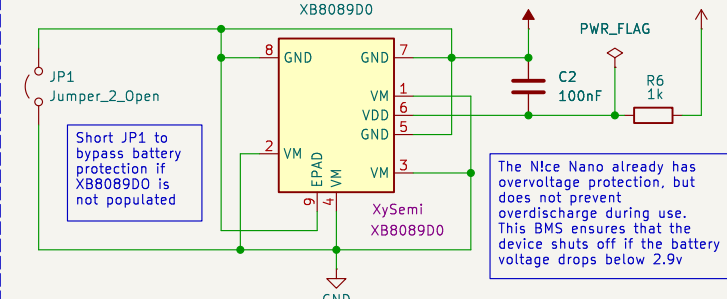
EXTRA GPIO



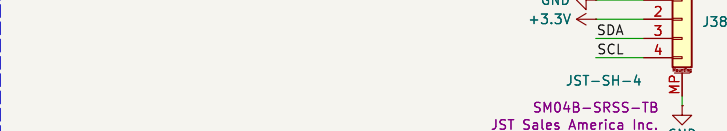
Nice!Nano nrf52840 microcontroller (or promicro)



BMS for overdischarge protection

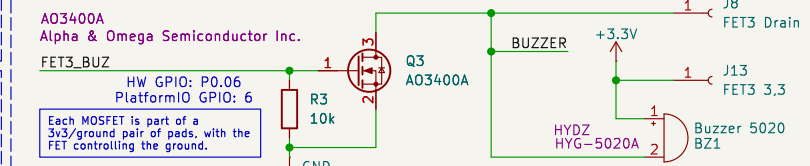
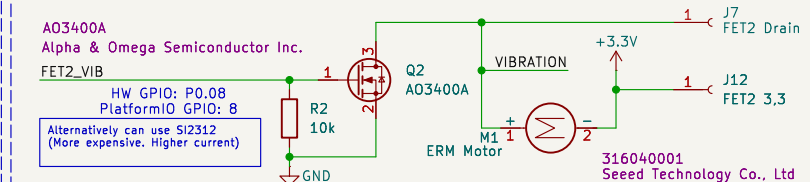
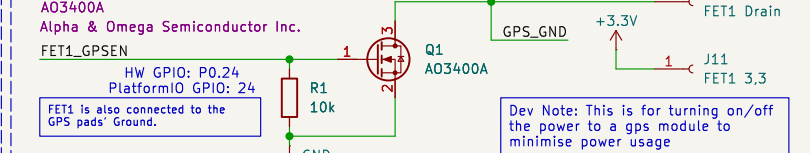


I2C Port (For extra sensors etc...)

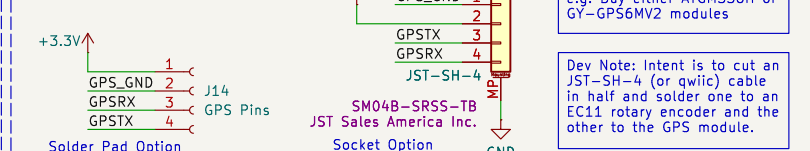


The JST-SH buddies
<https://hackaday.com/2022/05/04/the-connector-zoo-i2c-ecosystems>

Mosfets used for switching high current external perpherial



GPS Module



Development Source: https://github.com/mofosyne/roTec_pcb
 Fork Of FakeTec: https://github.com/jtljac/fakeTec_pcb
 Fork Of FakeTec: https://github.com/gargomoma/fakeTec_pcb/

Sheet: /
 File: fakeTecv5.kicad_sch

Title: roTec Rev5.3

Size: A4
 KiCad E.D.A. 9.0.6

Date: 2025-12-26

Rev: 5.3

Id: 1/1